More Buyers, More Competition? Reassessing Farm Product Procurement Markets in Developing Economies

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Abstract

Whether smallholder farmers benefit from market liberalization depends on competitive conditions in farm-product procurement. Development economics research routinely infers these conditions from regional buyer counts or the visible presence of multiple traders. Such structural proxies can mislead. In local procurement markets, repeated interaction, observable offers, and credible social sanctions allow buyer groups to coordinate and sustain collusion. Antitrust rarely disciplines this conduct in developing-country agriculture: the sector is often excluded from competition statutes, and where covered, enforcement concentrates on seller rather than buyer power. We identify three measurement pitfalls: treating regional buyer counts as local competition, equating multiple village-level traders with competitive bidding, and assuming competitive conditions are static. Fresh apple procurement in central China provides an illustration. A survey of 549 growers paired with interviews of the 15 leading traders shows that the average village is visited by only 3.3 of them, and home-town concentration is roughly four times the random-matching rate. Cold storage, a costly decision to delay sale, tracks farmers' own assessment of buyer rivalry rather than the observed buyer count: a one-point drop in perceived buyer rivalry raises storage probability by 19 percentage points, while the buyer count has no predictive power. Among storers, 81 percent describe storage explicitly as a bargaining tool against buyers. Pro-market reforms and entry alone do not ensure competitive procurement; structural proxies must be complemented with conduct-sensitive measures at the geographic and temporal scale of actual transactions.

Keywords: Buyer power, Oligopsony, Competition measurement, Agricultural markets, Market structure and conduct

JEL Classifications: O12, L13, Q12

Highlights:

- Headcounts mislead competition measurement in smallholder farm procurement.
- Village markets can sustain coordinated buyer conduct despite many traders.
- Three pitfalls confound regional, headcount, and static competition proxies.
- 1-pt rise in perceived buyer rivalry cuts storage probability 19 pp; N does not.
- Competition policy and research should focus on buyer conduct, not counts.

1 Introduction

Many developing countries have pursued pro-market reforms in recent decades, such as dismantling state marketing boards, liberalizing agricultural trade, and removing administered pricing. Although the benefits of such reforms are well understood, they have exposed smallholder farmers to prices and incomes determined by market forces. A fundamental premise of such reforms is that market-driven processes will generate competitive pricing in which farmers receive full value for their production. The promise is not fulfilled, however, if farmers are exposed to persistent buyer power in unregulated procurement markets (Rogers and Sexton, 1994; Sexton, 2000; Sexton and Lavoie, 2001; Macchiavello et al., 2022).

As commodities move from farms to village markets and onward through regional supply chains, intermediary traders play a key role in acquiring, aggregating, and transporting production from smallholder farms (Liverpool-Tasie et al., 2020). Yet the competitive conditions under which these traders operate remain poorly understood. Reardon (2015) characterizes the intermediary segments of agri-food value chains as the "hidden middle," largely neglected in mainstream academic literature and policy debates. Barrett et al. (2022) note that most theories of agricultural development and structural transformation abstract away from the intermediation roles of value chains.

To address this gap and to better understand market processes in developing-country settings, Bellemare (2022) argues that a "marriage" is needed between the fields of industrial organization and economic development. Macchiavello et al. (2022) make a complementary call, arguing that the tools of empirical industrial organization, from trigger-pricing models to relational-contracting frameworks, should be applied systematically to developing-country food value chains, with a particular gap on the buyer side of procurement from smallholder farmers. We heed this call by using an industrial organization framework to evaluate buyer competition in farm-product procurement markets in developing-country settings.

We argue that the common belief and modeling assumption that these markets are perfectly competitive may well be erroneous. Observation of many traders operating within a

producing region, i.e., "large N," can lead analysts to this erroneous conclusion given the relative ease of buyer collusion in these settings.

Despite strong policy interest in farmer welfare and market competition, rigorous analysis of competitive conditions in procurement from smallholder farmers has been rare due to lack of micro-level data on trader conduct, pricing strategies, and the local organization of procurement (Bellemare et al., 2022). Nonetheless, a growing body of research looks beyond buyer headcounts to examine trader conduct, often finding significant buyer power (Bergquist and Dinerstein, 2020; Kopp and Sexton, 2021).

In the absence of more comprehensive market data, researchers and policymakers have relied on indirect proxies, most commonly the number of traders operating in an area, to gauge market competition. Such a metric is attractive because it is observable and intuitive, with more buyers interpreted to mean greater competition. This presumption runs through much of the development literature. Reviewing 26 studies of sub-Saharan African food-crop markets, the influential synthesis by Dillon and Dambro (2017) concludes that evidence of non-competitive pricing in African food-crop markets is rare and that insufficient competition among traders is the exception rather than the rule. Consistent with this orientation, competitive or non-collusive benchmarks are adopted as a starting point across reduced-form studies of trader behavior (Fafchamps et al., 2005; Fafchamps and Hill, 2008), quantitative spatial models emphasizing transport costs and geographic segmentation (Porteous, 2020; Chatterjee, 2023), and general equilibrium frameworks (Cao and Birchenall, 2013; Davis and Hashimoto, 2024). Empirical work that directly observes village-level trading often reinforces the presumption by equating large buyer counts with effective rivalry (Pokhrel and Thapa, 2007; Sitko and Jayne, 2014; Do Nascimento Miguel, 2024), with high intermediary margins attributed to costs or managerial rents rather than coordinated conduct (Dawe et al., 2008).

We argue that inferences based on casual observation of trader numbers can substantially misrepresent the effective competitive environment faced by smallholder producers because these markets are well suited to support collusive behavior by buyers. Village procurement

markets typically feature repeated interaction among a stable set of traders who know one another personally and who may share common backgrounds or ethnicities. These settings can also feature low-cost monitoring of offers and transactions, and credible informal punishment mechanisms (Casaburi and Macchiavello, 2015), conditions that closely align with the environments in which collusion is sustainable, enforced by the credible threat of retaliation in response to suspected cheating on a collusive agreement. We also take up directly the standard objection that low barriers to entry should defeat any such coordination, showing that the channels through which entry could discipline cartels, including absorption as subcontractors, social exclusion, and annually renegotiated territorial allocations, are largely neutralized in our setting.

Building on these insights, we identify three specific pitfalls that affect beliefs regarding procurement from smallholder farmers. The first is that regional buyer counts represent a good measure of competitive conditions in local markets. While trader counts can measure potential competition, farmers in reality often receive few offers due to allocation of market territories among traders. The second is that increasing numbers of village-level traders imply more competitive bidding. Such an inference fails when conduct is coordinated through collusive mechanisms. The third is that competitive conditions are static rather than time-varying. Although collusive equilibria can be sustained in a repeated game context, as Porter (1983a,b) and Green and Porter (1984) demonstrated, collusive regimes may feature periods of heightened competition in the form of "punishment" inflicted in response to suspected cheating on the collusive agreement.

We develop these arguments conceptually and also draw on empirical evidence from fieldwork in China's fresh apple industry. We show that buyer groups in central China coordinate procurement through reference-price systems, employ sequential negotiation tactics, and enforce informal territorial boundaries that allocate relatively large numbers of traders operating in a procurement region into village-level territories. This allocation is visible in our buyer-side interview round. Although 15 leading traders (an estimated 70 percent of

annual procurement volume) are active across the study area, the average village is visited by only 3.3 of them; their visits concentrate in home townships at roughly four times the random-matching rate, and seven of the 69 study villages are left unvisited at harvest. We also exploit farmer storage decisions to illustrate farmers' response to perceived market competition. Farmers who perceive weak competition at harvest have stronger incentives to store (if they have access to such facilities) and sell later when competition may be more intense.

Using survey data from 549 apple growers, we find that a one-point increase in farmers' perceived buyer rivalry reduces the storage probability by 19 percentage points, while the farmer-reported buyer count has no detectable effect. Among storage adopters, 81 percent describe storage explicitly as a bargaining tool against buyers. While the econometric analysis does not directly test for buyer coordination, the divergence between what the structural proxy, N, predicts and what farmer behavior reveals matches the measurement failure that the qualitative evidence documents.

The analysis has implications for both competition policy and how researchers measure market power. Because antitrust rarely disciplines buyer conduct in developing-country agriculture, policy has to build alternative sources of farmer bargaining power: explicit coverage of agricultural procurement in competition statutes, support for farmer collective marketing, targeted interventions such as price floors when buyer power is documented, and investment in storage infrastructure that gives farmers a temporal outside option against harvest-period coordination. On the measurement side, the results suggest that competition should be assessed at the geographic and temporal scale at which transactions actually occur and that competition in procurement should be treated as a hypothesis to be tested rather than one that is maintained.

The remainder of the paper is organized as follows. Section 2 reviews antitrust enforcement in developing countries, establishing that legal constraints on buyer coordination are largely absent. Section 3 develops the conceptual framework: it establishes the prerequisites for a sustained buyer cartel, the mechanisms through which coordination operates, and

the three measurement pitfalls that follow. Section 4 presents qualitative evidence and a complementary quantitative illustration from China's apple procurement markets. Section 5 discusses policy implications, and Section 6 concludes.

2 Antitrust Enforcement in Developing Countries

Competition law has expanded widely across the developing world. More than half of developing countries now have antitrust statutes, up from fewer than 20 in 1989 (Waked, 2016; OECD, 2025). This proliferation reflects both endogenous forces, including political change and economic reform within developing countries, and external pressure, including prerequisites for WTO accession and conditions attached to World Bank and IMF lending (Pittman, 2025).

Adoption, however, does not imply effective enforcement. Competition law, in developing and developed economies alike, has focused almost entirely on seller rather than buyer power: cartels, price-fixing, and abuse of dominance by firms selling to consumers. Buyer-side coordination falls largely outside both statutory targets and enforcement practice. What enforcement does occur remains institutionally weak and does little to discipline market power in practice (Rodriguez and Menon, 2016). In Brazil and Egypt, merger control was absent for years and continues to be applied narrowly even after legal reforms; authorities rarely block transactions or impose meaningful remedies, enabling dominant firms to entrench inherited industrial concentration (Salgado, 2025). Evidence from India reveals a similar pattern of extreme under-enforcement. Despite the review of over a thousand mergers in highly concentrated sectors, virtually none have been prohibited (Gupta and Chhabra, 2025). Political prioritization further shapes outcomes. Huang et al. (2025) show, using the Alibaba case, that enforcement intensity in China reflects shifting state objectives more than systematic regulatory capacity. In Central and Eastern Europe, competition agencies have struggled to deter persistent cartel behavior by trade associations, relying primarily on weak fines and

informal pressure rather than resource-intensive prosecution (Pecotić Kaufman and Bernatt, 2025).

The problem of ineffective enforcement is even more pronounced in agricultural supply chains. Many developing countries exclude agriculture from their antitrust systems, albeit to varying degrees (Kirkwood, 2012; Balisacan, 2022), as Western counterparts did several decades earlier (Chauve et al., 2014).

Beyond weak public enforcement, a further and largely underappreciated limitation of competition regimes in developing countries is the near absence of private enforcement. Although many jurisdictions formally permit private parties to initiate competition lawsuits or seek damages, such actions remain exceedingly rare and frequently institutionally infeasible. Even in jurisdictions with relatively established private enforcement regimes, International Competition Network (2019) documents persistent challenges around proving infringement, quantifying harm, designing workable collective redress mechanisms, and building judicial expertise in competition matters, challenges that are likely to be even more binding in developing-country contexts. In fragmented agricultural markets characterized by dispersed smallholders and repeated interactions among itinerant traders, this enforcement gap is particularly important, weakening deterrence against buyer collusion.

Taken together, the weakness of both public and private antitrust enforcement in developing-country agriculture means that competitive outcomes in procurement markets are determined largely by the conduct of market participants themselves, rather than by legal or regulatory constraints.

3 Local Collusion and Measurement Pitfalls of Buyer Competition

Given the relative paucity of legal means to address buyer collusion in developing-country settings, the question turns to how traders operating in local procurement markets actually

behave and whether market forces can represent an effective deterrent to collusive behavior. This section examines, first, evidence that buyer coordination in these markets is more prevalent than the literature's competitive default assumes; second, the prerequisites for a sustained buyer cartel that the IO literature identifies, and the extent to which developing-country procurement from smallholder farmers satisfies them; third, the specific mechanisms through which coordination operates; and finally, how these mechanisms generate the three measurement pitfalls identified in the Introduction.

3.1 Documented Buyer Coordination in Agricultural Procurement

Recent empirical work challenges the competitive-default presumption by documenting coordinated conduct, persistent buyer power, and structural barriers that insulate traders from rivalry. Macchiavello et al. (2022) place collusion and cartels among the industrial-organization concepts most under-applied to food value chains, noting that empirical evidence on buyer-side coordination in developing-country procurement remains thin. An early example is Osborne (2005), who documented systematic markdowns in Ethiopian grain source markets despite the presence of multiple wholesalers; these markdowns furthermore appear only at harvest and substantially weaken in the lean season, a pattern he reads as evidence of seasonally enforced collusion. Bergquist and Dinerstein (2020) used field experiments in Kenyan agricultural markets to provide some of the clearest evidence of intermediary collusion. They showed that traders' earned median markups of 39 percent and captured roughly 82 percent of total market surplus, behavior consistent with joint profit maximization, not competition. Among their key findings was that induced entry had negligible effects on prices, suggesting that increasing the number of buyers does not meaningfully erode market power when conduct adjusts endogenously.

Other studies highlight how entry and market integration may expand participation without eliminating market power when traders are embedded in segmented networks or face mobility constraints. Following food market liberalization in Madagascar, Barrett (1997) shows

that trader numbers rose substantially, yet competition remained limited because entrants were confined to narrow niches determined by social identity and network position.

Infrastructure improvements can also alter outcomes without producing frictionless competition. Casaburi et al. (2013) find that rural road investments in Sierra Leone reshaped trader access and arbitrage opportunities in ways that suggest imperfectly competitive intermediation, while Masters (2008) emphasized that intermediary power can emerge endogenously from informational and bargaining advantages rather than from formal concentration. Structural approaches reinforce this picture: Macchiavello and Morjaria (2021) and Bulte et al. (2024) show that competition interacts with trust, enforcement limitations, and quality investment in non-linear ways, so that simple competitive benchmarks poorly approximate actual procurement behavior in imperfect institutional environments.

3.2 Prerequisites for a Sustainable Buyer Cartel

The industrial organization literature on cartel stability identifies specific conditions under which collusion can be sustained (Stigler, 1964; Green and Porter, 1984; Porter, 1983a,b). Synthesizing these conditions and adapting them to the buyer side, we identify eight prerequisites and argue that developing-country procurement from smallholder farmers satisfies each.

1. **High effective buyer concentration at the transactional margin.** Theoretical models of cartel stability presume concentration among the firms party to the agreement. The relevant concentration is transactional rather than regional: few buyers actually bid for any given parcel of output. Regional trader counts often overstate village-level contestability in developing-country procurement (Barrett, 1997; Casaburi et al., 2013).

2. **Stable production volumes and limited within-grade differentiation.** Stability on both the supply side (predictable annual volumes) and the demand side (standard-

ized grading) lowers the informational costs of coordination. Staple and tree crops typically exhibit predictable annual production, and standardized grade specifications allow incumbent buyers to align on a reference price for each grade.

3. **Frequent, small transactions among many dispersed sellers.** No single transaction is large relative to the market, so the short-run gain from undercutting any one farmer is small and the cost of detecting and punishing a deviation is correspondingly low. Smallholder agriculture in developing countries exhibits exactly this transaction structure (Barrett, 2008).

4. **Inelastic short-run supply at harvest.** When farmers cannot easily withhold output, buyers can sustain prices below the competitive level without triggering a supply response that would destabilize coordination. In most developing-country settings cold storage is costly or unavailable, so most smallholders must sell at or near harvest.

5. **Low-cost communication and coordination infrastructure among buyers.** Digital messaging, shared lodgings during trading seasons, kinship ties among local intermediaries, and informal trader meetings supply the monitoring and coordination channels that cartel stability requires.

6. **Credible punishment through exclusion from the buyer group.** Exclusion from shared logistics, reference-price information, storage access, and future territorial allocations supplies a swift, targeted, and bounded sanction. Normann et al. (2015) show experimentally that such a targeted-punishment channel sustains substantially more collusive pricing even among otherwise unrelated firms.

7. **Weak antitrust enforcement.** Buyer coordination is not legally actionable in the settings we study. Section 2 establishes this at length for developing-country agriculture.

8. **Meaningful barriers to effective entry.** Cartel coordination is undermined by new

buyers who decline to participate, so a stable cartel requires that entry be costly or effectively constrained. Agricultural procurement appears to violate this condition, since the headline physical asset of a trader (a truck) is cheap and non-sunk. We argue below that effective entry in fact requires relational assets that are slow to accumulate and visible to incumbents, and that incumbents routinely absorb, subcontract, or exclude would-be entrants who lack those assets.

Why entry does not defeat these cartels. Effective farm-gate procurement depends on considerably more than physical capital. Multi-season relationships with farmers, cultivar-specific quality-assessment skills, access to cold-storage facilities (some of which are controlled by incumbents), downstream distribution channels, and kinship or co-ethnic ties that let the buyer interpret local market signals are each slow to accumulate and visible during accumulation, so incumbents can detect genuine entry attempts well before they mature and coordinate a response.

Most non-local entrants find it more profitable to join the cartel than to compete with it. Faced with an informational disadvantage relative to local intermediaries, outsiders often subcontract to local agents operating under fixed fee schedules and informally integrated into incumbent buyer groups' shared logistics; entry of this kind converts would-be rivals into revenue-enhancing extensions of the existing cartel. Non-local buyers also typically serve distinct downstream channels, so competing at the sourcing stage yields product their distribution network cannot absorb efficiently, lowering the expected return on entry even before any cartel response. Bergquist and Dinerstein (2020) document a comparable dynamic in Kenyan agricultural markets: induced entry has negligible effects on prices, with trader conduct consistent with joint profit maximization.

Entrants who refuse absorption face credible discipline. Social exclusion from shared price-information channels, storage arrangements, and future territorial allocations supplies a targeted-punishment channel implemented through social ties rather than institutional

rules and therefore longer-lived and harder to defect against. Where territorial allocations are renegotiated annually, an entrant who competes aggressively in year t faces credible denial of access in year $t + 1$. And when new entrants are themselves small-scale local traders functioning as subcontractors for larger incumbents, adding more of them is not adding independent bidders at all: each additional small trader can deepen, rather than erode, buyer power by adding a further layer of intermediation and reinforcing territorial segmentation.

Together, these channels make the number of traders observed endogenous to the cartel's entry-management strategy; counting traders without classifying their conduct mistakes the outcome for the phenomenon. Mérel and Sexton (2017) show theoretically that downstream consolidation can coexist with, and even stimulate, upstream entry without eliminating buyer power, reinforcing that buyer presence is distinct from competitive conduct.

Large N is not a test of the prerequisites list. Standard oligopoly theory predicts that cartel stability is decreasing in the number of participants (Stigler, 1964), so that "large N" should signal competition even if the individual prerequisites are satisfied. This reasoning, however, assumes anonymous interaction among firms whose only connection is the market itself. In developing-country procurement, traders are typically embedded in social networks that alter the cooperation calculus: buyers who share regional origins, lodge together during trading seasons, or maintain long-standing personal relationships face social costs of defection that extend well beyond forgone market profits (Fafchamps, 2004). Historical evidence from trader coalitions demonstrates that shared social identity can sustain cooperative outcomes among groups far larger than anonymous-agent models predict as viable (Greif, 1993), and Haucap and Heldman (2023), analyzing fifteen German cartels prosecuted over two decades, find that participants were overwhelmingly drawn from homogeneous social networks, with industry associations providing the coordination platform and face-to-face meetings the primary decision-making venue. These cartels sustained themselves under one

of the world's most active competition regimes; in developing-country procurement, where enforcement against buyer conduct is largely absent, the same social fabric is unlikely to support less coordination. N is therefore uninformative on two counts: it fails to measure conduct directly, and the theoretical channel by which larger N should erode collusive discipline is weakened in the social environments where developing-country procurement takes place.

3.3 How Buyer Collusion Operates

Textbook expositions depict cartels as unstable: at least one participant always has a unilateral incentive to defect. This conclusion holds only in static or finite-period dynamic settings with perfect information, where cartel arrangements are generally not Nash or subgame perfect equilibria.

Collusive agreements can be sustained in infinite horizon game settings or in finite games with incomplete information. This result for infinite horizon games follows as a consequence of the Folk Theorem (e.g., Tirole (1989)), while the conclusion for finite-horizon games of incomplete information is due to players' incentives to communicate and acquire information, as first established by Kreps et al. (1982). Green and Porter (1984) established that in industries subject to demand uncertainty cartel agreements could involve periods of reversion to more competitive pricing in response to negative demand shocks and declining prices.[1] Sexton and Lavoie (2001) catalogue the empirical implementation of these trigger-pricing models in US agricultural processing industries, showing that even in well-functioning competition regimes the static-competitive benchmark can be rejected for specific commodity markets, a baseline that should be treated as more permissive, not less, in settings lacking comparable antitrust oversight.

Sustaining such collusive equilibria in practice requires repeated interaction, monitoring,

[1]Such behavior is consistent with Nash equilibrium because declining price could also signal some cartel members cheating on the agreement, requiring firms to respond by expanding output as a punishment strategy.

and credible punishment (Green and Porter, 1984), conditions unusually well satisfied in village procurement, where the same traders return each season, transactions are visible and prices travel quickly through social networks, and exclusion from a buyer group or sourcing territory is a credible sanction. Controlled experimental evidence isolates the mechanism. Normann et al. (2015) show that firms able to organize as a *buyer group* with the power to exclude defecting members sustain substantially more collusive pricing than the same firms without such exclusion power, even when each firm acts independently in the output market. Village procurement satisfies the experimental precondition, credible intra-group exclusion, through a richer channel than a lab can construct: shared social identity, common logistics, and annually renegotiated territorial access. If the mechanism operates among anonymous experimental subjects, it should operate more forcefully among traders whose exclusion costs persist across seasons.

Against this backdrop, three mechanisms are particularly important. The first is **reference-price coordination**. Traders align on an offer schedule for grades and qualities, limiting undercutting, with coordination facilitated by rapid communication and reinforced by punishment threats. The trader survey in Mitra et al. (2018) is consistent with this mechanism: 43 to 51 percent of traders admitted exchanging information about recent price offers with other traders before making offers to farmers, and about 40 percent said they checked recent sale prices with farmers. The same signalling channel has been identified in wholesale benchmarks in US agricultural markets (Richards, 2022), where it operates despite Robinson-Patman and related statutes constraining price discrimination and coordination, suggesting that reference-price coordination is a general-purpose cartel technology rather than an artifact of a particular institutional environment.

The second is **territorial allocation and non-aggression norms**. Traders respect village boundaries or "informal ownership" of sourcing areas, reducing rivalry not by lowering the total number of traders in a region but by shrinking the number who contest any given village at a given time. A village can thus remain effectively "thin" even when the

surrounding region is "thick." A particularly clear case is tomato marketing in Ghana, where Robinson and Kolavalli (2010) document how organized trader leadership ("market queens") controls distribution networks, allocates trucks between sourcing areas, and requires negotiation for downward farm-gate price adjustments, an explicit coordination infrastructure that limits both entry and competitive bidding.

The third is **control of farmer switching**. Relational arrangements such as credit, repeated dealing, and interlinked services raise the cost to farmers of switching buyers, dampening competitive pressure even when multiple traders are present. In Sierra Leone cocoa markets, Casaburi and Reed (2022) describe credit as a tool to "lock in" supply, preventing other traders from competing for a farmer's output. Kopp et al. (2024) demonstrate a related mechanism in Indian okra production: production contracts, which bind farmers more tightly to a single buyer, are associated with systematically greater markdowns than marketing contracts, suggesting that the form of vertical coordination directly shapes the degree of buyer power exercised over smallholders.

Together, these instruments generate a key conceptual wedge: the relevant object is not the number of buyers observed in the region, but whether farmers face competitive bidding at harvest.

3.4 From Collusive Conduct to Measurement Failure

Pitfall 1: Regional buyer counts reflect local competition. A common approach is to infer competitiveness from the number of buyers operating in a county or prefecture, on the implicit assumption that regional buyer availability translates into village-level contestability. Under territorial allocation, however, the effective set of buyers who can credibly bid in a particular village at harvest can be small and stable even when the regional buyer count is large. Defining the "right" market boundary is itself problematic, because the boundary is endogenous to the very coordination mechanisms that the number-of-buyers proxy is meant

to capture.[2] For policymakers evaluating market access programs, this mismatch means that increasing regional trader numbers may not translate into better prices at the village level, and that program evaluations relying on regional headcounts may overstate welfare gains.

Pitfall 2: Multiple traders present ensure competition. A second common inference is that observed multiplicity at the village level implies competition. This reasoning collapses structure and conduct, violating the foundational IO distinction between them. Multiple traders can be present yet coordinate through reference prices, territorial norms, or sequential negotiation tactics, leaving the village "crowded" in appearance while effective rivalry is muted. This measurement failure is not merely theoretical: Bergquist and Dinerstein (2020) find experimentally that induced entry into Kenyan agricultural markets has negligible effects on prices, consistent with traders acting as joint profit maximizers. Structural proxies, whether Herfindahl–Hirschman index (HHI) calculations, concentration ratios, or simple headcounts, are therefore unreliable without direct evidence on conduct.[3] Impact evaluations that use trader presence as a measure of competitiveness may consequently overstate the competitive effects of entry-promoting interventions.

Pitfall 3: Competitive conditions are static rather than time-varying. Standard structural proxies treat competition as a fixed characteristic of a market: measure the number of buyers or a concentration index at a point in time, and the result is taken to characterize the competitive environment. The repeated-game conditions of village-level procurement can sustain collusive outcomes as equilibrium behavior, but collusion need not be permanently stable. Collusive arrangements are subject to disruption by outside entry, demand shocks,

[2]Casaburi and Reed (2022) estimate through a Cournot model that the effective number of competitors for a given farmer's supply in Sierra Leone cocoa markets is roughly 50 percent higher than the village-level trader count, implying that narrow boundaries can understate competition. The inference does not generalize, however: in settings where inter-village coordination or territorial norms restrict effective entry, even a large regional pool need not translate into contestable local procurement.

[3]Alternatives such as residual supply estimation can measure oligopsony power without structural proxies (Durham and Sexton, 1992), but they require detailed price and quantity data rarely available in developing-country settings.

changes in downstream market conditions, and shifts in the composition of buyer groups, so effective competition can fluctuate over time while the number of traders remains unchanged (Green and Porter, 1984). A buyer headcount observed at any single point captures none of this variation. The same ten traders can correspond to vigorous competition during a collusion breakdown or to suppressed rivalry when coordination holds. The static treatment of competitive conditions thus introduces a temporal dimension of measurement failure that compounds the spatial (Pitfall 1) and conduct-level (Pitfall 2) problems identified above. Monitoring programs that assess competitive conditions at a single point in time may capture an unrepresentative episode, leading to incorrect assessments of the need for policy intervention.

4 Evidence from China's Apple Procurement Markets

The conceptual framework in Section 3 identifies mechanisms through which buyer coordination can decouple structural proxies from effective competition. This section presents evidence from China's fresh apple industry that illustrates these mechanisms in practice. We begin with qualitative evidence from fieldwork documenting how buyer groups coordinate procurement, then complement this with a quantitative illustration showing that farmers' costly economic decisions align with the conduct-based framework: storage behavior responds to perceived competition but not to buyer headcounts.

4.1 Setting

Yanchang County is located in a semi-arid plateau region of Shaanxi Province, central China, where late-maturing Fuji apples account for 85–90 percent of total production. The county comprises over 260 villages across eight rural townships, with farm-gate prices ranging from 2.5 to 12 yuan per kilogram depending on quality, timing, and buyer competition. Although over one hundred itinerant traders travel to the county each harvest season, procure-

ment is organized through buyer groups that coordinate internally while competing across groups. We draw on our fieldwork conducted during the 2023 and 2024 harvest seasons and an original survey of 549 smallholder apple growers conducted during the 2024 harvest (October–December 2024). Households were selected through stratified random sampling, with proportions across townships determined by the county agricultural bureau to reflect the distribution and density of apple growers.

4.2 Qualitative Evidence: Buyer Coordination in Practice

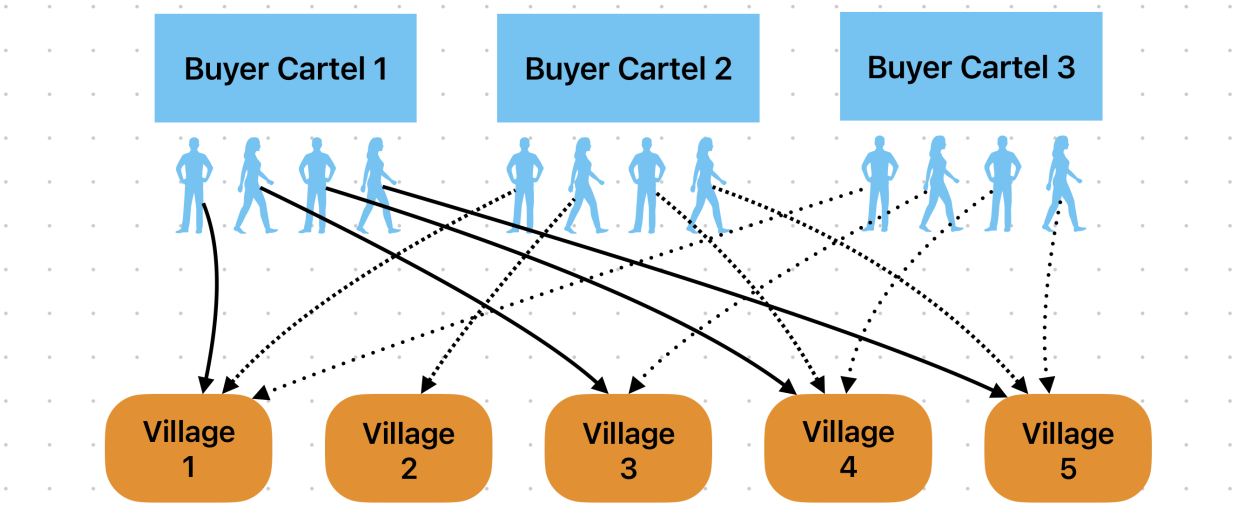
The qualitative findings reported below draw on semi-structured interviews with approximately 30 traders, cold storage operators, and local agricultural officials; participant observation at village-level procurement sites and county trading venues; and informal conversations at cold storage facilities where traders congregated during the 2023 and 2024 harvest seasons. Patterns were triangulated across multiple independent informants and directly observed during fieldwork.

Within each buyer group, members align purchasing strategies before entering villages, share real-time price information through WeChat, and employ sequential negotiation tactics to manipulate farmer expectations. In one documented pattern, a first trader visits a village and offers a price without closing any deals; a colleague follows the next day with a deliberately lower offer, prompting farmers to accept rather than risk losing a buyer entirely. Such tactics closely resemble the price-signaling and bid-rotation strategies identified in formal cartel enforcement, yet they operate informally and without legal consequence in rural procurement settings where antitrust oversight is absent.

Territorial allocation reinforces these pricing arrangements. Traders within a buyer group establish informal procurement territories at the village level and respect one another's sourcing areas within a crop year. As one local trader described: "I will not step into a village if another member of my group has already contacted it during the same crop year." The trader's account maps cleanly onto the closed-buyer-group equilibrium identified experi-

mentally by Normann et al. (2015), with one strengthening: the exclusion power that the experiment imposes by design is here socially constructed and longer-lived. The threat of being excluded from the group, and with it access to shared WeChat channels, storage arrangements, and future territorial allocations, is the sanction that sustains the territorial norm. These boundaries are renegotiated annually, so that individual villages often face only one or two traders from a given group at any point during harvest, even though the broader county hosts dozens of active buyers (Figure 1).

Figure 1: Market Territory Allocation Among and Within Buyer Groups in Yanchang County



Note: Schematic illustration based on fieldwork observations. Each buyer group coordinates internal territory assignments, while inter-group territories overlap in high-quality sourcing areas. Villages served by only one buyer group face effective monopsony.

The stability of intra-group coordination rests on a simple economic logic: members of a buyer group typically serve distinct downstream distribution channels, so that competing at the sourcing stage would raise procurement costs without expanding market share. As one trader explained: "We avoid competition during the sourcing stage because our downstream distribution channels are almost entirely distinct from one another." This channel segmentation makes coordination self-enforcing: the short-run gains from undercutting are small because poaching a fellow group member's supply yields apples that the deviator's own distribution network may not absorb efficiently.

Between groups, however, rivalry can be intense. The manager of a local aggregator in Yanchang County observed: "The competition between field-buyer cartels is fierce. They are wary of other cartels capturing higher-quality apples or encroaching on their supply bases, which could undermine their market position. Consequently, cartels are generally unwilling to collaborate with one another." The coexistence of intra-group coordination and inter-group rivalry is a classic pattern in industrial organization: it implies that the relevant unit of competitive analysis is the buyer group, not the individual trader, and that counting individual traders overstates the number of independent decision-makers in the market.

Non-local traders, who travel from distant provinces, exhibit analogous coordination: during the initial trading period they commonly lodge in the same hotels in the county seat, where they form interest groups and align procurement strategies before dispersing to villages. This pattern, face-to-face alignment at a neutral venue followed by decentralized implementation through phone and messaging, is the same procedure Haucap and Heldman (2023) document for prosecuted German cartels, whose executives used hotels at industry fairs to set cartel terms before dispersing to individual transactions. The Chinese counterpart operates without legal consequence: what required subterfuge, and subsequent prosecution, in Germany is routine practice at the county trading venues we observed.

Non-local traders face an informational disadvantage relative to local intermediaries, who possess detailed knowledge of village-level supply conditions, quality variation, and cold storage logistics. Rather than competing head-to-head, many subcontract local agents (*daiban*) to source apples on their behalf. *Daiban* operate under two arrangements: "full agents" handle sourcing, sorting, and transportation, earning approximately 0.70 RMB/kg; "assistant agents" focus solely on matchmaking, earning roughly 0.08 RMB/kg. They function as informal extensions of larger buying operations, adding a layer of intermediation rather than rivalry, so outside entry replicates the same coordination dynamics at a different scale rather than intensifying competition.

The magnitude of the gap between regional buyer counts and village-level reality is strik-

ing: although 100 to 150 buyers may appear within Yanchang County over the course of the marketing year, they are dispersed across more than 260 villages, so that fewer than one buyer is present per village on average. Effective competition is determined at the village level, yet the structural proxy is typically measured at the county level. Buyer presence at any given village also fluctuates substantially from year to year, driven by idiosyncratic factors such as trader route planning, yield expectations, and shifting inter-trader relationships. As two growers from Luozishan Township reported: "Seven or eight traders came to our village two years ago; last year only two showed up; this year we have no idea." A local trader confirmed that his routes change annually depending on "information about yields, quality, and my relations with other traders," and that once he secures enough volume he skips remaining villages entirely. Once apples enter cold storage, competition also shifts from the village to the storage site, so buyer rivalry varies not only in intensity but in location.

The relationship between market concentration and farmer welfare is counterintuitive. In most townships of Yanchang County, small-scale traders function as intermediaries for larger buyers rather than as independent competitors, and increasing their number can deepen buyer power by adding intermediation and reinforcing territorial segmentation. Angou Town is the exception: three large buyers dominate procurement, the township government facilitates collective negotiation, and oversight is feasible because the buyers maintain permanent local offices. As one Angou village secretary put it: "The government helps farmers negotiate collectively with the three major buyers. Oversight of these companies is also more straightforward." The policy objective should not be maximizing buyer headcounts per se, but ensuring that the buyers present engage in genuinely independent bidding, whether their number is large or small.

The Angou case is the exception rather than the rule. In most townships, village-level regulation of trader conduct has proven practically infeasible. The Party Secretary of Nanpo Village, interviewed during the first week of the 2024 harvest, captured the constraint succinctly: "I cannot demand or prevent traders from colluding; all I can do is plead with them

for a slightly higher farm-gate price for my village. Otherwise, they'll simply leave and source apples from other villages. They have countless options elsewhere." This powerlessness at the village level reinforces the argument in Section 2: absent formal antitrust enforcement, competitive outcomes are determined by the conduct of traders themselves, and the conduct that emerges is shaped by coordination incentives that village authorities can neither monitor nor sanction.

4.3 Quantitative Evidence

The exercise that follows is illustrative, not a causal test of coordinated conduct: it shows that the structural proxies most often used to gauge competition, such as the number of buyers and village-level concentration indices, fail to track the outcomes and decisions that hinge on rivalry at the point of transaction, while conduct-sensitive measures predict them. The evidence addresses Pitfalls 1 and 2; the temporal pitfall enters as a further reason the static buyer-count proxy is uninformative rather than as a separate test.

Figure 2 previews the argument in three panels before any regression. Panel (A) plots the village-level mean farmgate price against the number of distinct buyers visiting the village: the OLS slope is essentially zero, a first visual indication that buyer counts do not track the prices farmers receive (the second pitfall). Panel (B) contrasts the regional buyer count (15) with the distribution of village-level counts, whose mean is only 3.34, a roughly five-fold gap consistent with the territorial allocation documented qualitatively in the preceding subsection (the first pitfall). Panel (C) reports mean highest prices by sale month with 95% confidence intervals; prices rise from October to December even though the number of buyers farmers encounter does not move monotonically with this gradient, providing the within-season illustration that is folded into the second pitfall in Section 4.3.3. The remainder of this subsection unpacks each of these patterns.

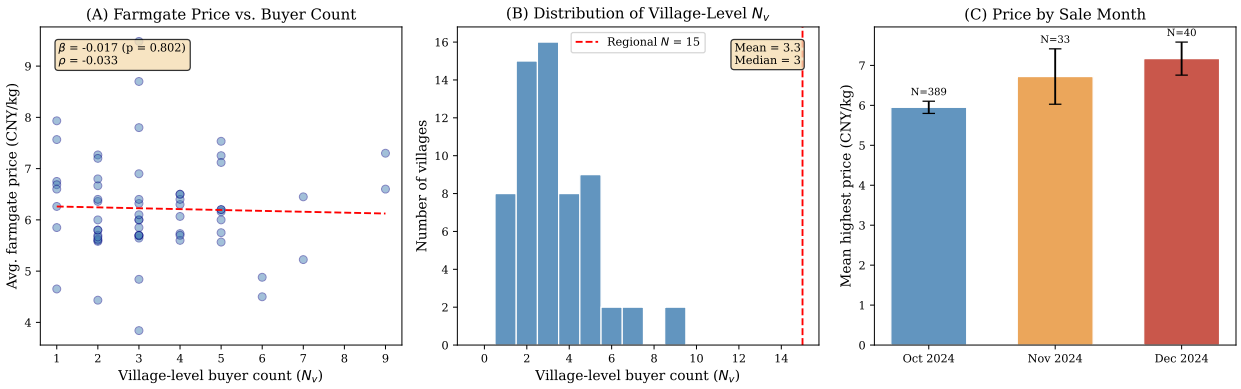


Figure 2: Three Patterns in the Raw Data

Notes: Panel (A): village-level scatter of mean farmgate price against N_v, the number of distinct buyers visiting each village; the dashed line is the OLS fit with slope -0.017 ($p = 0.789$). Panel (B): histogram of village-level buyer counts across the 62 villages in the buyer-visit dataset, with the regional total (15) marked by the vertical dashed line. Panel (C): mean highest farmgate price by sale month with 95% confidence intervals; sample sizes are annotated above each bar.

4.3.1 Data

The farmer survey was administered with Institutional Review Board approval between October 10 and December 13, 2024, in collaboration with the Yanchang County Women's Federation and the county Agricultural Bureau. We used one-layer stratified random sampling across the eight townships, with proportions (30%, 15%, 15%, 15%, 10%, 5%, 5%, 5%) set by the Agricultural Bureau to reflect the distribution and density of apple growers. Eligibility required cultivating no more than 50 mu (about 8.2 acres) and deriving at least 90 percent of agricultural income from Fuji production; interviews were conducted with household heads or, when they were elderly, with the primary labor-force member. The analytic sample consists of 549 households across 69 villages in 8 towns (Table 1).[4] Two hundred of these farmers (36.4%) used cold storage during the marketing season: 33 in self-built facilities, 49 in rented small-scale commercial storage, and 118 in rented large-scale commercial storage. Storage rental costs are 0.4–0.6 CNY/kg, roughly 5–15 percent of the harvest price, making storage a non-trivial intertemporal commitment.

[4]Five village names recur across different townships, so a naive deduplication on village name alone undercounts the universe by five. All village-level computations in this section identify villages by the (town, village) pair.

Table 1: Summary Statistics

Variable	N	Mean	SD	Min	Max
Storage adoption (=1)	549	0.364	0.482	0.00	1.00
Buyer count (farmer-reported)	549	3.377	2.145	0.00	9.00
Subjective competition (1–5)	549	3.007	0.752	1.00	5.00
Highest farmgate price (CNY/kg)	549	6.116	1.537	1.60	13.40
Prior-year price (CNY/kg)	463	5.955	1.481	2.00	11.00
Liquidity constrained (=1)	549	0.158	0.366	0.00	1.00
Risk aversion (CRRA)	549	0.262	0.441	-0.20	0.90
Farm size (mu)	549	12.398	8.269	1.00	98.00
Age (years)	549	59.078	9.747	17.00	83.00
Household size	549	2.628	1.142	0.00	7.00
Household labor	549	1.654	0.684	0.00	4.00
Storage as bargaining tool (=1)	549	0.479	0.500	0.00	1.00
Wider marketing purpose (=1)	549	0.510	0.500	0.00	1.00
Late sale (Nov/Dec, =1)	549	0.133	0.340	0.00	1.00

Notes: Sample of 549 apple farmers in Yanchang County, Shaanxi Province, 2024 marketing season. Subjective competition is self-reported on a 1 (very weak) to 5 (very strong) scale. CRRA is derived from a Holt-Laury lottery game. Prior-year price has 463 non-missing observations.

Three blocks of variables are central. *Storage behavior* is measured by a binary indicator for cold-storage use at harvest, together with storage type and a binary indicator for late sale in November or December. *Competition measures* include the farmer's subjective assessment of buyer rivalry at harvest, recorded on a 1 (very weak) to 5 (very strong) scale, and an objective village-level buyer count constructed from the buyer-side survey described below. Following Macchiavello and Morjaria (2021), who use the number of coffee mills within a 10 km radius as a concentration metric in Rwanda's coffee chain, we define the village-level buyer count N_v as the number of distinct traders visiting village v at harvest. The survey also asks farmers whether they store apples as a bargaining tool against buyers and whether they store to access wider marketing opportunities beyond the harvest-period buyer pool. *Controls* include risk aversion, elicited both through a self-reported five-point scale (Dohmen et al., 2011) and through a Holt–Laury-type lottery game (Holt and Laury, 2002) from which we back out a constant relative risk aversion (CRRA) coefficient; a binary liquidity-constraint indicator defined as in Albuquerque et al. (2024) and Stephens and Barrett (2011), equal

to one if the farmer reports both difficulty accessing credit and an urgent need to sell at harvest; distance to the nearest storage facility; farm size in mu; the prior-year farmgate price; and standard demographics including age, education, household labor, and a social-capital indicator for whether a household member has ever served as village leader.

The farmer survey is paired with a buyer-side interview round conducted in December 2024. We interviewed the 15 leading traders operating in the study area during the 2024 October harvest, who together are estimated to account for roughly 70 percent of annual procurement volume and include the five most influential local intermediaries. For each we elicited the specific villages visited, the cold-storage facilities from which they had sourced during the preceding two months, and each buyer's downstream wholesale selling price. This yields a bipartite buyer–village network of 207 directed visit links across 62 of the 69 study villages; the remaining seven villages in the farmer survey received no visits from the 15 leading traders during the October harvest. Village-level farmgate prices, wholesale prices, and the implied wholesale–farmgate margin are derived by merging the farmer-reported and buyer-reported data; the margin can be computed for the 62 villages present in both datasets.

The principal timing variable is the farmer-reported sale month. October sales ($N = 389$) are at-harvest spot transactions with a storage rate of 10.3%. November ($N = 33$) and December ($N = 40$) sales are storage-mediated by construction (storage rate 100%). Mean highest prices rise monotonically from 5.95 to 6.72 to 7.17 CNY/kg across these months and anchor the within-season analysis in Section 4.3.3.

4.3.2 Regional Versus Local Buyer Counts

The first measurement pitfall asserts that regional buyer counts overstate village-level competition when traders allocate territories. The buyer-side interview round allows a direct test. The 15 leading traders we interviewed, who together are estimated to account for roughly 70 percent of annual procurement volume, operated across the study area during the 2024 harvest; yet the mean village was visited by only $\bar{N}_v = 3.34$ (SD = 1.84, median

= 3). Nearly two-thirds of villages (62.9%) hosted three or fewer buyers, and over one-third (37.1%) hosted two or fewer (Table 2). The regional count overstates the effective village-level count by a factor of roughly five.

Table 2: Village-Level Buyer Presence: Regional vs. Local Competition

Statistic	Value
Regional buyer count	15
Villages in buyer network	62
Village-level effective N_v	
Mean	3.34
Standard deviation	1.84
Median	3
Range	1–9
Share with $N_v \leq 3$	62.9%
Share with $N_v \leq 2$	37.1%
Buyer territory types	
Singleton (home-town only / 2–3 towns / 4+ towns)	6
Intermediate (home-town only / 2–3 towns / 4+ towns)	0
Regional (home-town only / 2–3 towns / 4+ towns)	9
Mean home-town visit share	61.9%

Notes: Village-level buyer counts (N_v) constructed from a census of 15 buyers reporting the specific villages they visited during the 2024 harvest. Singleton buyers operate exclusively within their home town; regional buyers span four or more towns. Home-town visit share is the fraction of each buyer's village visits located within their home town.

Table A.1 traces this gap to territorial specialization. Six of the fifteen buyers (B09–B12, B14, B15) operated exclusively in their home town, each with a home-town visit share of 100 percent. The remaining nine spanned four or more towns but still concentrated procurement near home; the mean across buyers of the home-town visit share is 61.9 percent. This pattern is difficult to rationalize with random or purely demand-driven spatial procurement.

Permutation test. Under the null that buyer–village matches are independent of buyer identity, we shuffle home-town labels across buyers 10,000 times, holding each village's total visit count fixed, and recompute the pooled share of visits that buyers make to their own home towns. The null distribution has mean 13.6% and standard deviation 5.2%. The observed pooled home-town share is 50.7%, far in the right tail of the null (Figure A.1,

$p < 0.0001$). The null of random matching is rejected decisively.

Exclusivity regression. We also estimate a linear probability model for whether buyer b visits village v:

$$\Pr(\text{Visit}_{bv} = 1) = \beta_0 + \beta_1 \cdot \text{SameTown}_{bv} + \beta_2 \cdot \text{VillageSize}_v + \beta_3 \cdot \text{OtherBuyers}_v + \varepsilon_{bv}, \quad (1)$$

where SameTown_{bv} equals one when buyer b's home town matches village v's town, VillageSize_v is the number of surveyed farmers in village v, and OtherBuyers_v counts other buyers visiting village v. Distance data at finer resolution are unavailable; the same-town indicator captures the primary territorial dimension.

Table 3: Exclusivity Regression: Determinants of Buyer–Village Visit

	Coefficient	Std. Error
Constant	0.0269	(0.0238)
Same town	0.7785***	(0.0295)
Village size (farmers)	0.0007	(0.0012)
Other buyers in village	0.0293***	(0.0070)
Observations	930	
R^2	0.3897	

Notes: Linear probability model. Dependent variable: $\mathbb{1}$(buyer b visits village v). Robust (HC1) standard errors in parentheses. Distance controls not available; the same-town indicator captures the primary territorial dimension. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

The same-town coefficient is 0.778 ($p < 0.001$): sharing a home town raises the probability of a visit by 78 percentage points, and it dominates every other predictor. OtherBuyers enters with a small positive coefficient (0.029, $p < 0.001$), consistent with demand-driven clustering toward larger or more accessible villages, but its magnitude is an order of magnitude smaller than the home-town effect.

Farmer-level corroboration. On the farmer side, 28 of the 349 non-storers (8.0%) cite pre-booking (*guoshang yuding*, "the buyer has already booked me") as the reason for selling

at harvest, a farmer-level footprint of the same territorial allocation, concentrated in the towns where fieldwork documented the tightest intra-group norms.

Together, these results align with the qualitative account of informal territorial allocation among buyer groups. They are difficult to reconcile with an environment in which the regional headcount reliably gauges local competition.

4.3.3 Buyer Counts and Effective Rivalry

The second measurement pitfall asserts that multiple traders operating in a village need not imply competitive bidding when their conduct is coordinated. We evaluate this from two directions: whether structural proxies predict prices and margins, and whether farmers' own costly decisions respond to the structural count in the direction that a competitive-procurement model would require.

Prices and margins do not respond to the structural count. Table 4 reports village-level OLS regressions of the mean farmgate price, and of the wholesale–farmgate margin, on N_v and on the Herfindahl–Hirschman index (HHI) constructed from within-village visit shares. The coefficient on N_v is small, negative, and insignificant in every specification: $\hat{\beta} = -0.017$ ($p = 0.789$) in the bivariate model and $\hat{\beta} = -0.032$ ($p = 0.634$) with town fixed effects and village-size controls. HHI performs no better ($\hat{\beta} = 0.447$, $p = 0.293$). The null persists when the outcome is the margin (column 4). Village-level margin dispersion is itself large: the mean margin is 0.87 CNY/kg with standard deviation 1.04 and range from -1.87 to $+3.33$, and 12 of the 62 villages (19.4%) show negative margins, farmgate prices that exceed reported wholesale prices (Figure A.2). An ANOVA rejects equal mean margins across towns ($F = 2.72$, $p = 0.017$), but 73.9% of the total variance is within-town. That is hard to square with a competitive intermediation benchmark in which transport, handling, and sorting costs should vary principally across towns rather than within them.

At the farmer level ($N = 549$), a town-fixed-effects regression yields $\hat{\beta} = 0.079$ ($p =$

Table 4: Village-Level Regressions: Farmgate Price and Margin on Buyer Count

	Avg. Farmgate Price			Margin
	(1)	(2)	(3)	(4)
N_v (buyer count)	-0.017	-0.032		-0.033
	(0.064)	(0.067)		(0.073)
HHI			0.447	
			(0.424)	
Village size	No	Yes	Yes	Yes
Town fixed effects	No	Yes	Yes	Yes
Observations	62	62	62	62
R^2	0.001	0.264	0.274	0.270

Notes: OLS regressions at the village level. Dependent variable in columns (1)–(3) is the average farmgate price (CNY/kg) from the farmer survey; column (4) uses the wholesale–farmgate margin. N_v is the number of distinct buyers visiting the village, constructed from the buyer census. HHI is computed from buyer visit shares. Robust (HC1) standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

0.005), reflecting village-level confounders. Adding village fixed effects absorbs this heterogeneity: the within-village coefficient is $\hat{\beta} = 0.231$ ($p = 0.430$), a noisily estimated zero. Within village, one additional buyer has no detectable relationship with the price a farmer realizes.

Storage as revealed preference. Farmer storage decisions provide a revealed-preference test that is difficult to rationalize under competitive procurement. Storage is costly: rental fees of 0.4–0.6 CNY/kg, spoilage risk, and forgone liquidity. If the harvest-period farmgate market were competitive, farmers would store only when the expected intertemporal price appreciation covers these costs; the decision would be governed by intertemporal arbitrage and would not respond to rivalry at harvest. If the market is instead non-competitive, storage takes on a second role as an outside option against the coordinated buyer group. We estimate a logit of storage adoption:

$$\Pr(\text{Store}_i = 1) = \Lambda(\alpha + \beta_1 N_i + \beta_2 \text{Comp}_i + \mathbf{X}_i'\boldsymbol{\gamma} + \text{TownFE}), \tag{2}$$

where N_i is the farmer-reported buyer count at harvest, Comp_i is the 1–5 subjective competition assessment, and $\mathbf{X}_i$ includes the CRRA coefficient, the liquidity-constraint indicator, distance to storage, farm size, and the prior-year farmgate price.

Table 5: Storage Adoption Decision: Logit Estimates and Average Marginal Effects

	(1) Structure	(2) Conduct	(3) Full	AME
Buyer count	-0.064		0.329***	0.035***
	(0.055)		(0.117)	(0.012)
Subjective competition (1–5)		-0.693***	-1.474***	-0.158***
		(0.168)	(0.331)	(0.032)
Liquidity constrained			-0.198	-0.021
			(0.390)	(0.042)
Risk aversion (CRRA)			-0.901**	-0.097***
			(0.359)	(0.037)
Far from storage			0.097	0.010
			(0.487)	(0.052)
Farm size (mu)			0.102***	0.011***
			(0.021)	(0.002)
Prior-year price			0.540***	0.058***
			(0.130)	(0.013)
Town FE	Yes	Yes	Yes	Yes
Observations	463	463	463	463
Log-likelihood	-201.2	-192.4	-155.6	
Pseudo R^2	0.222	0.256	0.398	

Notes: Logit coefficients in columns (1)–(3); average marginal effects (AMEs) from Model (3) in the rightmost column. Dependent variable: storage adoption (=1). Subjective competition is self-reported on a 1 (very weak) to 5 (very strong) scale. Robust standard errors in parentheses. Town fixed effects included in all specifications. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Two patterns stand out in Table 5. First, subjective rivalry predicts storage, and structural rivalry does not. In the structure-only specification (Model 1), N_i has no significant effect ($\hat{\beta}_1 = -0.064$, $p = 0.242$). In the conduct-only specification (Model 2), subjective competition is a strong negative predictor ($\hat{\beta}_2 = -0.693$, $p < 0.001$). In the full specification (Model 3), subjective competition dominates, with an average marginal effect of -0.191 ($p < 0.001$): a one-point increase in perceived rivalry reduces the storage probability by 19.1 percentage points. The structural buyer count does not recover in the expected direction; its

coefficient in Model 3 is positive and significant ($\hat{\beta} = 0.329$, $p = 0.005$), so that conditional on perceived rivalry, farmers in higher-N villages are, if anything, more likely to store. We read this not as direct evidence of coordinated conduct but as a further reason N is an unreliable summary of effective competition. Control coefficients behave as expected: risk-averse farmers store less, larger farms store more, and higher prior-year prices predict storage.

Second, farmers themselves describe storage as a bargaining instrument. Among the 200 storers, 80.5% report using storage explicitly as a bargaining tool, and 87.5% report using it to access wider marketing opportunities beyond the harvest-period buyer pool. The two motives are not mutually exclusive, but the 80.5% volunteering bargaining as a purpose is difficult to reconcile with a harvest-period bid that already reflects competitive conditions.

Cross-town evidence. The town-level cross-section reinforces the picture (Table 6, Figure A.3). Under competitive procurement, towns with the most buyers should store the least. The relationship is non-monotonic: Town G (Leichi), with $\bar{N} = 2.35$ and the highest at-harvest price (7.49 CNY/kg), stores at zero percent; Town C (Zhangjiatan), with $\bar{N} = 3.73$ and a lower at-harvest price, stores at 74.4%. The at-harvest price, a direct summary of conduct, determines the storage margin; the buyer count does not.

Table 6: Town-Level Cross-Section: Storage, Buyer Counts, and Prices

Town	Name	N	Mean Buyers	Storage Rate	Mean Price	Subj. Comp.
G	Leichi	48	2.35	0.0%	7.49	2.85
F	Luozishan	67	2.00	3.0%	4.95	2.73
E	Zhengzhuang	26	1.96	7.7%	6.43	2.58
D	Heijiabu	73	2.84	8.2%	5.48	2.99
H	Qilicun	14	5.00	21.4%	7.39	3.57
A	Jiaokou	87	3.62	41.4%	6.15	3.05
B	Angou	74	4.96	43.2%	6.63	3.45
C	Zhangjiatan	160	3.73	74.4%	6.06	2.98

Notes: Each row summarizes farmer-level data aggregated to the town level. Mean buyers is the average of farmer-reported buyer counts. Storage rate is the share of farmers adopting cold storage. Mean price is the average highest farmgate price received (CNY/kg). Subjective competition is measured on a 1–5 scale.

Within-season timing. The within-season price trajectory offers a further illustration that the static buyer count fails to track rivalry (Table A.2 in the appendix). Mean highest prices rise from 5.95 CNY/kg in October to 6.72 in November and 7.17 in December (premia of 13 and 20 percent over the at-harvest mean), yet the buyer-count pattern is not even monotone: November sellers report *fewer* buyers than October sellers (2.58 vs. 3.12) and December sellers report more (5.00). Whatever the price gradient tracks, it is not contemporaneous N. We do not read the wedge as a buyer-power rent: the average net payoff to storage across adopters was close to zero after rental and spoilage costs, with positive realized returns concentrated among well-timed sellers.

Across the pieces, the pattern is coherent: buyer counts and HHI fail to predict prices, margins, or storage in the direction a competitive-procurement model requires, while perceived rivalry does, and farmers' own stated reasons for storage align with the buyer-power reading. Nothing here proves coordinated conduct on its own. Read alongside the qualitative evidence of the preceding subsection, however, these patterns match the pitfall framework of Section 3 and are difficult to rationalize under the competitive default much of the development literature maintains. The Angou Town contrast reinforces the same reading from the opposite direction: three regulated buyers deliver better outcomes than the many coordinated ones that prevail elsewhere in the county.

5 Policy Implications

The stakes of accurate competition measurement are high: whether smallholder farmers capture the gains from market liberalization, infrastructure investment, and value chain development depends on whether the procurement markets they face are genuinely competitive. Studies routinely use the number of traders, market concentration indices (four-firm concentration ratio CR4, HHI), or observed marketing margins to characterize competitive conditions (Hernandez and Torero, 2013; Dillon and Dambro, 2017). As World Bank (2025)

and Clapp et al. (2025) emphasize, these proxies fail to capture the mechanisms through which concentration translates into power over farmers. If competition is assumed rather than verified, assessments of liberalization, infrastructure investment, and trader entry programs may overstate welfare gains; and interventions that genuinely disrupt coordination may be undervalued against a competitive baseline.

Our analysis motivates several specific policy directions. First, **enactment and enforcement of competition laws** that explicitly cover agricultural procurement. Where laws exist, enforcement agencies rarely investigate buyer conduct in procurement markets, in part because the conventional wisdom treats these markets as competitive. Experience from high-income jurisdictions shows that applying antitrust tools to agricultural procurement is far from automatic (Richards, 2022); explicit attention to the design of competition law in agricultural contexts is therefore needed alongside enactment.

Second, **support for farmer collective marketing** can directly counter buyer power by aggregating supply and improving farmers' outside options. Cooperatives, producer organizations, and collective bargaining arrangements shift the structure of negotiation from bilateral transactions between individual farmers and coordinated buyer groups toward more balanced exchanges. The effectiveness of collective action depends critically on institutional design, enforcement of cooperative governance, and the capacity to sustain participation (Barrett, 2008), but the potential to mitigate buyer power is well established in theory and supported by varied empirical evidence.

Third, **price floors or government procurement programs**, though often viewed unfavorably in competitive market frameworks, may improve welfare when buyer power is present. A price floor that would introduce deadweight loss in a competitive market can raise total surplus when it binds above the collusive price but below the competitive price (Sexton and Lavoie, 2001; Sexton, 2013).

Fourth, **investment in storage infrastructure** for storable crops can strengthen farmers' bargaining position by introducing a dynamic dimension to competition. Storage lets

farmers exploit temporal disruptions to coordination: rather than selling into a harvest-period market where coordination is at its strongest, farmers who can store shift sales to periods when competition may be more intense, new buyers enter, or downstream demand improves. Our survey evidence supports this logic: 81 percent of storage users report using storage explicitly as a bargaining instrument against buyers. Public investment in accessible, affordable cold storage facilities can thus serve as a competitive instrument, expanding the set of farmers who possess outside options against coordinated buyer groups and raising the cost to cartels of sustaining depressed harvest-period prices.

6 Conclusion

This paper challenges a pervasive default in development economics: the presumption that the number of buyers in agricultural procurement markets provides a reliable indicator of competition. We have argued that this presumption conflates market structure with market conduct, overlooking the conditions under which buyer coordination can flourish in smallholder supply chains. The institutional environments that characterize village-level procurement, including repeated interaction among a stable set of traders, high observability of transactions, and credible informal punishment mechanisms, are those in which collusive or quasi-collusive equilibria are sustainable under standard repeated-game theory. When buyer conduct softens rivalry, headcounts, concentration ratios, and observed margins cease to be informative about the competition farmers actually face. Yet even where collusion is sustainable it need not be permanently stable: the same market can alternate between coordinated and competitive phases in response to outside entry, demand shocks, or shifts in buyer-group composition, adding a temporal dimension of measurement failure that static proxies cannot capture.

We documented these dynamics through three lenses: a review of competition-law enforcement, showing that formal deterrents against buyer collusion are largely absent in

developing-country agriculture; fieldwork in China's apple procurement markets, documenting reference-price systems, territorial allocation, and sequential negotiation tactics among buyer groups; and an original farmer survey paired with interviews of the 15 leading traders, in which the average village is visited by only 3.3 of these buyers, and farmers' storage decisions respond strongly to perceived buyer rivalry (a 19-percentage-point drop per point of the subjective rating) but not to the buyer count itself. The econometric evidence does not directly test for coordination, yet the pattern mirrors the measurement failure the qualitative fieldwork documents.

For policymakers, the overarching message is that private procurement cannot be presumed competitive: building farmer bargaining power requires institutional investment rather than reliance on market liberalization alone.

For researchers studying market outcomes in procurement from smallholder farmers, three lessons emerge. First, competition should be measured at the geographic and temporal scale at which transactions actually occur, not at administrative boundaries. Second, structural proxies should be complemented with conduct-sensitive indicators: subjective farmer assessments, behavioral responses like storage or search, and direct observation of trader interactions. Third, treating competition as the null hypothesis in procurement settings where repeated interaction, monitoring, and punishment mechanisms are present is a strong and often untested maintained assumption that should be explicitly justified.

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Appendix A. Supporting Exhibits

Table A.1: Buyer Characteristics and Territorial Behavior

Buyer	Home Town	Villages	Towns	Home Share	Wholesale	Type
B01	H (Qilicun)	18	7	11%	6.5	Regional
B02	H (Qilicun)	17	6	12%	8.0	Regional
B03	H (Qilicun)	17	6	12%	7.5	Regional
B04	C (Zhangjiatan)	19	6	58%	6.0	Regional
B05	C (Zhangjiatan)	19	6	63%	8.0	Regional
B06	A (Jiaokou)	16	5	50%	6.5	Regional
B07	B (Angou)	17	4	24%	6.5	Regional
B08	F (Luozishan)	23	4	52%	7.0	Regional
B09	A (Jiaokou)	13	1	100%	7.5	Singleton
B10	A (Jiaokou)	12	1	100%	7.0	Singleton
B11	B (Angou)	6	1	100%	9.0	Singleton
B12	B (Angou)	5	1	100%	8.5	Singleton
B13	D (Heijiabu)	17	4	47%	6.5	Regional
B14	E (Zhengzhuang)	6	1	100%	6.0	Singleton
B15	G (Leichi)	2	1	100%	9.0	Singleton

Notes: Home share is the fraction of village visits within the buyer's home town. Wholesale price in CNY/kg. Singleton = operates in home town only; Intermediate = 2–3 towns; Regional = 4+ towns.

Table A.2: Prices, Storage, and Buyer Contact by Sale Month

Sale Month	N	Storage Rate	Mean Price	SD Price	Mean Buyers
October 2024	389	10.3%	5.95	1.54	3.12
November 2024	33	100.0%	6.72	2.03	2.58
December 2024	40	100.0%	7.17	1.33	5.00
Nov–Oct premium			+0.77		
Dec–Oct premium			+1.22		

Notes: October sales are predominantly at-harvest spot transactions (storage rate 10.3%); November and December sales are storage-mediated. Price in CNY/kg.

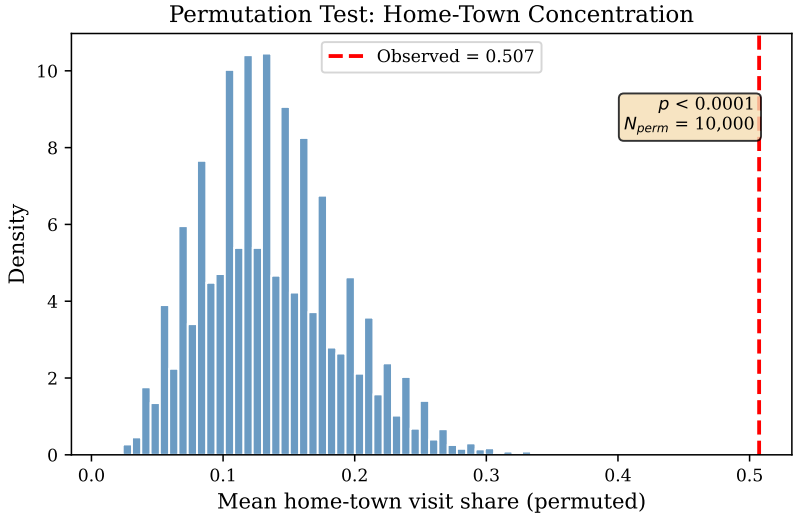


Figure A.1: Permutation Test: Home-Town Concentration of Buyer Visits

Notes: Distribution of pooled home-town visit share under 10,000 random permutations of buyer home-town labels holding village visit counts fixed. The dashed red line marks the observed share ($p < 0.0001$).

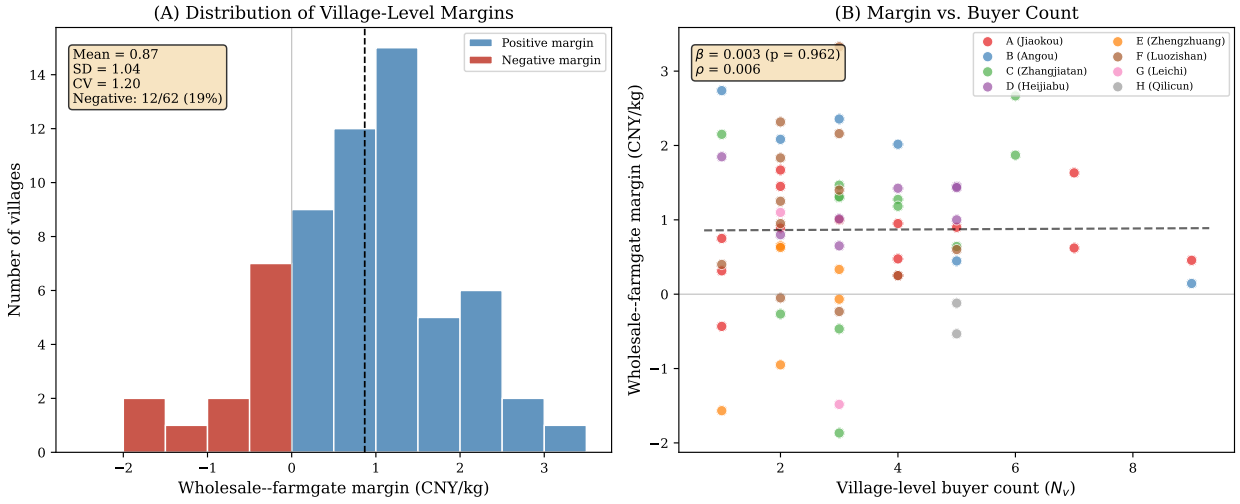


Figure A.2: Village-Level Marketing Margins

Notes: Panel (A): distribution of wholesale–farmgate margins across 62 villages; red bars indicate villages with negative margins. Panel (B): scatter of margins against the number of distinct buyers visiting each village, colored by town; the dashed line is the OLS fit. Margins are measured in CNY/kg.

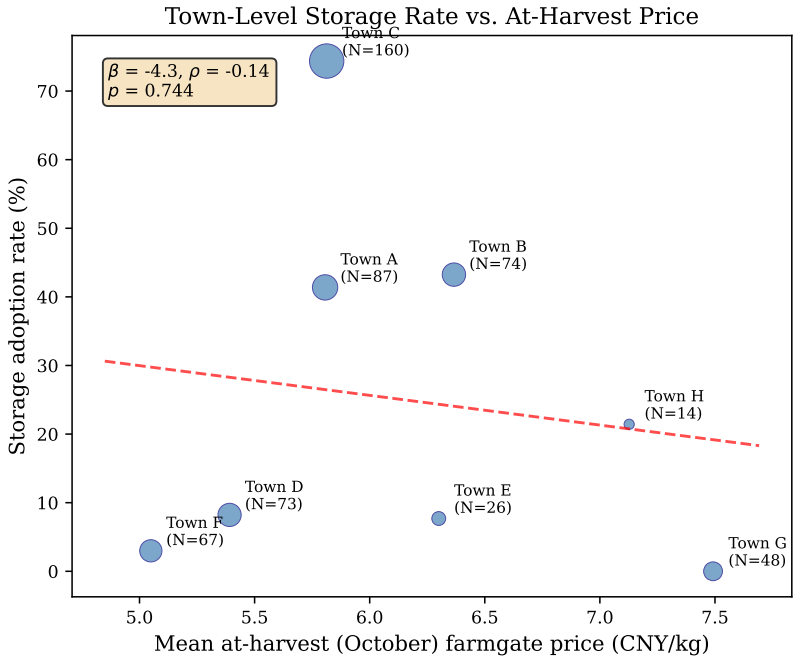


Figure A.3: Town-Level Storage Rate vs. At-Harvest Farmgate Price

Notes: Each point represents one of the eight towns; bubble size is proportional to the number of surveyed farmers. The horizontal axis is the mean highest farmgate price among farmers who sold at harvest (October). The dashed line is the OLS fit.